

Santiago del Palacio

Academic Formation

2014–2018 **PhD in Astronomy**, *Faculty of Astronomical and Geophysical Sciences, La Plata National University*, La Plata, *Thesis defense mark: 10/10*.

Completion date: Nov. 20, 2018

2008–2014 **Ms. S. in Astronomy**, *Faculty of Astronomical and Geophysical Sciences, La Plata National University*, La Plata, *GPA: 9.56/10*.

Programming Skills

- Proficient in bash, FORTRAN, Python and SQL.
- Software control using Github.
- Working knowledge of Machine Learning

Research visits

Sep–Nov 2019 **University of Southampton**, *Host: Dr. Diego Altamirano*.

Jun–Jul 2018 **University of Barcelona**, *Host: Dr. Valentí Bosch-Ramon*.

Jun–Jul 2016 **University of Barcelona**, *Host: Dr. Valentí Bosch-Ramon*.

Research experience

2022–2026 **AoP Postdoctoral Fellow (Chalmers University of Technology)**, *Topic: Astrophysical outflows and radio observations*.

2019–2022 **Postdoctoral CONICET Research Fellow (National Research Council)**, *Topic: Conditions for the generation of γ -ray emission in massive binaries*, Supervisor: Dr. Romero, Gustavo E.

2014–2019 **PhD CONICET Research Fellow (National Research Council)**, *Topic: Investigation of non-thermal processes in the vicinities of massive stars*, Advisor: Dr. Bosch-Ramon, Valentí, Co-advisor: Dr. Romero, Gustavo E.

2013–2014 **CIC training fellowship for university students**, Supervisor: Dr. Combi, Jorge.

2012–2013 **CIN fellowship for stimulating scientific vocation**, Advisor: Dr. Bosch-Ramon, Valentí, Co-advisor: Dr. Romero, Gustavo E.

Awards

2022 Third place poster prize in the Astronomdagarna 2022 (Sweden).

2020 *Carlos M. Varsavsky* award to the best PhD thesis in astronomy in the period 2018–2019 (Argentina).

2019 Fellowship from the COSPAR Capacity Building Fellowship Programme (ESA). Topic: research visit to the University of Southampton, UK. Amount: €570.

- 2015 Distinction to the best poster contribution in the I Annual Meeting of PhD Students of the Faculty of Astronomical and Geophysical Sciences (UNLP).
- 2014 Joaquín V. González Prize, Distinction awarded to students of the University of La Plata with the highest marks. La Plata City Council, Buenos Aires, Argentina.

Teaching Experience

Graduate Experience

- 2023–2025 **Invited Guest Lecturer on radio continuum emission in the course *Radiointerferometry***, Faculty of Astronomical and Geophysical Sciences, UNLP, La Plata.
- 2020–2022 **Head of Practical Assignments of the courses *Introduction to Relativistic Astrophysics* and *Introduction to Black-Hole Astrophysics***, Faculty of Astronomical and Geophysical Sciences, UNLP, La Plata.
- 2016–2020 **Graduate Teaching Assistant of the courses *Introduction to Relativistic Astrophysics* and *Introduction to Black-Hole Astrophysics***, Faculty of Astronomical and Geophysical Sciences, UNLP, La Plata.
- 2014–2020 **Graduate Teaching Assistant of Calculus**, Faculty of Engineering, UNLP, La Plata.
- 2015,2016 **Graduate Teaching Assistant of the Summer Entrance Course**, Faculty of Exact Sciences, UNLP, La Plata.

Undergraduate Experience

- 2011–2014 **Teaching Assistant of the Summer Entrance Course**, Faculty of Astronomical and Geophysical Sciences, UNLP, La Plata.
- 2012–2014 **Teaching Assistant of Algebra**, Faculty of Astronomical and Geophysical Sciences, UNLP, La Plata.
- 2011–2012 **Teaching Assistant of Calculus**, Faculty of Engineering, UNLP, La Plata.
- 2011 **Teaching Assistant of Calculus**, Faculty of Exact Sciences, UNLP, La Plata.

Teaching Formation

- 2017 **Seminar on Teaching Formation**, Faculty of Exact Sciences, UNLP, La Plata, duration: 12 h. Approved.
- 2015 **Graduate course: Geogebra didactic aspects for teaching basic science and engineering**, Faculty of Engineering, UNLP, La Plata, duration: 40 h. Mark: 10/10.
- 2014 **Workshop: Teaching and evaluation in university classrooms**, Faculty of Astronomical and Geophysical Sciences, UNLP, La Plata, duration: 12 h. Approved.

Observing Experience

Approved Proposals as PI

- 2025 **PI of JVLA Proposal**, *Constraining AGN coronal emission with the VLA*, Time: 2.6 h.
- 2025 **PI of ToO Swift Proposal**, *Swift-UVOT monitoring of the AGB star WHya quasi-simultaneous with an ALMA flaring episode*, Time: 0.6 ks.

- 2025 **PI of ToO Swift Proposal**, *Swift X-ray monitoring of HE0047–1756 quasi-simultaneous with a VLA observation*, Time: 8 ks.
- 2025 **PI of ToO Swift Proposal**, *Swift X-ray monitoring of Cyg X-3 simultaneous with a radio VLBI observation*, Time: 6 ks.
- 2024 **PI of JVLA Proposal**, *The corona hunt in high- z lensed quasars*, Time: 3.45 h.
- 2024 **PI of ALMA Proposal**, *Resolving the WR winds from the colliding-wind binary Apep*, Time: 1.2 h.
- 2024 **PI of NuSTAR Proposal**, *Testing the connection between neutrino and hard X-ray emission in AGNs*, Time: 60 ks.
- 2023 **PI of XMM-Newton Proposal**, *Searching for coronal activity in isolated AGB stars*, Time: 42 ks.
- 2023 **PI of ToO Swift Proposal**, *Coordinated X-ray and millimetre observations of an AGB star*, Time: 8 ks.
- 2022 **PI of NuSTAR Proposal**, *Evolution of the X-ray emission from the colliding-wind binary HD 93129A after its periastron passage*, Time: 60 ks.
- 2022 **PI of ToO Swift Proposal**, *Swift X-ray monitoring of Cyg X-3 simultaneous with a radio VLBI observation*, Time: 6 ks.
- 2021 **PI of Joint Chandra–NuSTAR Proposal**, *Evolution of the X-ray emission from the colliding-wind binary HD 93129A after its periastron passage*, Time: 60 ks with both instruments.
- 2019 **PI of XMM-Newton Proposal**, *Relativistic particle acceleration in the Wolf-Rayet bubble G2.4+1.4*, Time: 58 ks.
- 2017 **PI of Joint Chandra–NuSTAR Proposal**, *A unique opportunity to observe the extreme massive binary HD 93129A near periastron passage*, Time: 50 ks with both instruments.
- 2016 **PI of JVLA Proposal**, *Search of synchrotron radiation in two interacting stellar bubbles (2nd part)*, Time: 1.33 h.
- 2016 **PI of JVLA Proposal**, *Search of synchrotron radiation in two interacting stellar bubbles*, Time: 3.23 h.
- 2015 **PI of JVLA Proposal**, *Is the radiation from the brightest colliding-wind binaries polarized?*, Time: 7.9 h.
- 2015 **PI of JVLA DDT Proposal**, *Is the Emission from the Wind-Collision Region of WR146 Polarized?*, Time: 1 h.

Approved Proposals as Co-I

- 2026 **Co-I of ALMA Large Project Proposal**, *ChANCE: Characterising the AGB Near-Circumstellar Environment*, Time: 70.5 h.
- 2026 **Co-I of ALMA Proposal**, *A radio-submm investigation of obscured AGN and extreme nuclear starburst in local LIRGs*, Time: 14.6 h.
- 2026 **Co-I of VLBA Proposal**, *Deciphering the Mechanism of a TDE Re-brightening Using VLBA*, Time: 4 h.

- 2025 **Co-I of eMERLIN Proposal**, *Probing compact obscured nuclei at high-angular resolution with e-MERLIN*, Time: 57 h.
- 2025 **Co-I of eMERLIN Proposal**, *Resolving the OH megamaser emission in a complete northern sample of U/LIRG-CONs*, Time: 72 h.
- 2025 **Co-I of eMERLIN Proposal**, *Monitoring the Flaring Evolution of an Extraordinary TDE*, Time: 15.5 runs.
- 2025 **Co-I of NICER DDT Proposal**, *NICER X-ray monitoring of Cyg X-3 simultaneous with a radio VLBI observation*, Time: 7 ks.
- 2024 **Co-I of ALMA Proposal**, *The first 40 GHz ALMA continuum survey of nearby AGN*, Time: 29.5 h.
- 2024 **Co-I of ALMA Proposal**, *A novel approach to identify the most heavily obscured supermassive black holes in the Universe*, Time: 8.3 h.
- 2024 **Co-I of VLBA DDT Proposal**, *Imaging a flaring nuclear transient with VLBA*, Time: 10 h.
- 2024 **Co-I of JVLA DDT Proposal**, *Constraining the period of radio flares in a nuclear transient with the VLA*, Time: 5 h.
- 2024 **Co-I of JVLA DDT Proposal**, *Characterizing the accretion-ejection activities in a nuclear transient with VLA*, Time: 5 h.
- 2024 **Co-I of JVLA DDT Proposal**, *AT 2019avd: Tracking the evolution of its long-lived radio flare*, Time: 2 h.
- 2024 **Co-I of JVLA DDT Proposal**, *Characterizing the accretion-ejection activities in a nuclear transient with VLA*, Time: 5 h.
- 2024 **Co-I of JVLA DDT Proposal**, *AT 2019avd: Tracking the evolution of its long-lived radio flare*, Time: 2 h.
- 2023 **Co-I of JVLA DDT Proposal**, *AT 2019avd: Constraining origin of the bright radio flare revealed by VLASS*, Time: 2.5 h.
- 2023 **Co-I of uGMRT Proposal**, *Wolf-Rayet bubbles as potential particle accelerators*, Time: 12 h.
- 2022 **Co-I of NICER DDT Proposal**, *NICER X-ray monitoring of Cyg X-3 simultaneous with a radio VLBI observation*, Time: 7 ks.
- 2022 **Co-I of EVN Short Request**, *Verifying the goodness of a candidate calibrator near Cyg X-3*, Time: 2 h.
- 2022 **Co-I of EVN Proposal**, *Growing black holes or supernova factories in extremely opaque LIRGs*, Time: 40 h.
- 2020 **Co-I of ESA DS3 + CLTC-CONAE Proposal**, *Search for Periodic Radio Signals from the Gamma-ray Source Fermi J1913+0515 Associated to SS433*, Time: 48 h.
- 2020 **Co-I of GMRT Proposal**, *Investigating particle acceleration associated with putative single Wolf-Rayet stars*, Time: 18 h.
- 2018 **Co-I of GMRT Proposal**, *Investigating the low-frequency radio emission in Wolf-Rayet Bubbles*, Time: 16 h.

- 2017 **Co-I of JVLA Proposal**, *Mapping magnetic fields in the stellar bow shock EB27*, Time: 3 h.
- 2017 **Co-I of VLBA Proposal**, *Is the emission from the wind-collision region of WR 146 polarized?*, Time: 12 h.
- 2016 **Co-I of XMM-Newton Proposal**, *The massive binary HD 93129A: an extreme wind-wind collision during AO16*, Time: 60 ks.
- 2016 **Co-I of GMRT Proposal**, *Testing the non-thermal radio emission from the putative gamma-ray source WR 11*, Time: 24 h.
- 2016 **Co-I of GMRT Proposal**, *Spatial Distribution and Variability of the Non-Thermal Emission Associated with Proplyd-like Objects Near the Cygnus OB2 Region*, Time: 14 h.

Telegrams

- 2024 **Co-author of ATel #12482**, *Detection of a new giant glitch in the Vela Pulsar observed from the Argentine Institute of Radio astronomy*, PuMA Collaboration.
- 2024 **Co-author of ATel #16580**, *Detection of a new glitch in PSR J1048–5832 observed from the Argentine Institute of Radio astronomy*, PuMA Collaboration.
- 2022 **Co-author of ATel #15838**, *Detection of a new glitch in the pulsar PSR J1740–3015*, PuMA Collaboration.
- 2022 **Co-author of ATel #15638**, *Confirmation of glitch event observed in PSR J0742–2822 at the Argentine Institute of Radioastronomy (IAR)*, PuMA Collaboration.
- 2021 **Co-author of ATel #14806**, *A new Glitch in the Vela Pulsar (PSR B0833–45/PSR J0835–4510)*, PuMA Collaboration.
- 2019 **Co-author of ATel #12482**, *Radio observations following the recent glitch of Vela Pulsar (PSR B0833–45)*, PuMA Collaboration.
- 2018 **Corresponding author of ATel #12323**, *Follow up of the radio flare from the magnetar XTE J1810–197 at 1.4 GHz*, On behalf of the PuMA Collaboration.

Participation in Scientific Meetings

Invited Talks

- 2024 **Continuum emission from radio-quiet AGN**, *1st Nordic Meeting on Neutrino and Radio Astronomy*, Gothenburg.

Contributed Talks

- 2026 **Probing the Sizes of AGN Coronae with ALMA-B2**, *Unveiling ALMA Band 2 Workshop*, Bologna.
- 2026 **Coronal emission from radio-quiet AGNs at high redshift**, *3rd Sweden SKA Days*, Stockholm.
- 2025 **Non-thermal emission from colliding stellar winds**, *Variable Galactic Gamma-Ray Sources (VII)*, Barcelona.

- 2025 **Continuum mm emission from SMBH coronae in (dust-obscured) AGN**, *Behind a Curtain of Dust (V)*, Kiruna.
- 2024 **Radio emission from stellar bowshocks in massive runaway stars**, *2nd National SKA Science Day Sweden*, Gothenburg.
- 2024 **Modelling the millimeter emission from SMBH coronae in dust-obscured AGN**, *European Astronomical Society Annual Meeting (EAS2024)*, Padova.
- 2023 **Radio observations and modelling of non-thermal sources**, *52nd Young European Radio Astronomers Conference (YERAC)*.
- 2023 **Colliding-wind binaries**, *Variable Galactic Gamma-Ray Sources (VI)*, Innsbruck.
- 2022 **Non-thermal emission from colliding-wind binaries**, *7th International Symposium on High-Energy Gamma-Ray Astronomy (Gamma2022)*, Barcelona.
- 2022 **The physics behind the low-frequency emission from massive stars**, *AS-TRON/JIVE colloquium talks*.
- 2021 **Non-thermal radiation in the colliding-wind binary Apep**, *63th Annual Meeting of the Argentinian Astronomical Association*.
- 2020 **Varsavsky prize talk 2020: Non-thermal radiation associated to massive stars**, *62nd Annual Meeting of the Argentinian Astronomical Association*.
- 2019 **X-ray view of the extreme colliding-wind binary HD 93129A during periastron**, *New Results in X-ray Astronomy 2019 (MSSL, UK)*.
- 2018 **Pulsar monitoring from the IAR observatory**, *Mini-course on gravitational waves (FCAGLP)*.
- 2018 **Non-thermal emission from stellar bow shocks**, *Massive stars and Supernovae (Nidia Fest)*.
- 2017 **Multi-frequency study of the massive binary HD 93129A**, *III Annual Meeting of PhD Students of the Faculty of Astronomical and Geophysical Sciences (UNLP)*.
- 2017 **High-Energy Astrophysics**, *5th National Meeting of Astronomy Students*.
- 2017 **γ -ray emission in blazars produced by interactions between BLR clouds and jet**, *60th Annual Meeting of the Argentinian Astronomical Association*.
- 2015 **A model for the non-thermal emission of the very massive colliding-wind binary HD93129A**, *58th Annual Meeting of the Argentinian Astronomical Association*.
- 2014 **Interactions of relativistic particles in stellar winds**, *II Argentinian-Brazilian Meeting on Gravitation, Astrophysics, and Cosmology*.
- 2013 **Equivocando las preguntas: Métodos actuales de evaluación**, *1st National Meeting of Astronomy Students, Córdoba (Argentina)*.

Posters

- 2024 **Analysis of single pulses in pulsar data using AI**, *Mini-conference on AI for radioastronomy*, Gothenburg.
- 2022 **Non-thermal emission from the colliding-wind binary Apep**, *Astronomdagarna 2022*, Gothenburg.

- 2021 **The high-energy emission from the massive colliding-wind binary HD 93129A near periastron**, *237th meeting of the American Astronomical Society*.
- 2016 **Non-thermal Processes in systems with relativistic jets**, *II Annual Meeting of PhD Students of the Faculty of Astronomical and Geophysical Sciences of the UNLP*.
- 2016 **Gamma-ray predictions for the very massive colliding-wind binary HD 93129A**, *6th International Symposium on High-Energy Gamma-Ray Astronomy (Gamma2016)*.
- 2015 **Relativistic particle content in the most massive colliding-wind binary in the Galaxy**, *High-Energy Phenomena in Relativistic Outflows V*.
- 2015 **Investigación de procesos no-térmicos en las cercanías de estrellas de gran masa**, *I Annual Meeting of PhD Students of the Faculty of Astronomical and Geophysical Sciences of the UNLP*, awarded the best poster distinction.

Attendant

- 2024 **Space Array kick-off meeting**, Gothenburg.
- 2021 **International Cosmic Ray Conference 2021**, Berlin (virtual meeting).
- 2021 **European Astronomical Society Annual Meeting**, Leiden (virtual meeting).

Organization

- 2024 **Astronomdagarna 2024 (SOC)**, Lund.
- 2014 **2nd National Meeting of Astronomy Students**, La Plata.

Other Courses Attended

- 2019 **Equality & Diversity Briefing course**, *Southampton University*, UK, duration: 2 hs, successfully completed with a score of 100%.
- 2017 **Workshop ‘Latin American X-ray School for Different Astrophysical Scenarios: Data Analysis of the XMM-Newton, Chandra and NuSTAR missions’**, *COSPAR*, Viedma, duration: 2 weeks.

Formation of human resources

- 2022–2026 **Co-advisor of the Ph.D. in Astronomy of Ezequiel Zubieta (Advisor: Federico García)**, *Topic: Intensive monitoring of Southern pulsars*, UNLP, La Plata. Grade: 10/10
- 2021 **Advisor of the M. Sc. Thesis in Physics of Javier Martínez (Co-advisor: Gustavo E. Romero)**, *Topic: Investigation of bow shocks from hyper velocity stars*, UNLP, La Plata. Grade: 10/10.
- 2021 **Advisor of the M. Sc. Thesis in Astronomy of Agustina Blanco (Co-advisor: Paula Benaglia)**, *Topic: Radiointerferometric study of the colliding-wind binary WR 147*, UNLP, La Plata. Grade: 10/10.
- 2020 **Advisor of the M. Sc. Thesis in Astronomy of Sofía Valentina Sosa Fiscella (Co-advisor: Luciano Combi)**, *Topic: High-precision timing study of the radio pulsar J0437-4715 using IAR antennas*, UNLP, La Plata. Grade: 10/10.

Outreach activities

- 2026 Outreach talk at Slottsskogsobservatoriet (Gothenburg). Talk title: 'Neutron Stars and Pulsars'.
- 2025 Co-editing a press release for Chalmers University on 'a black hole system caught wobbling as it devours a star'.
- 2025 Lecturer in *Rymdforskarskolan* (Swedish National Space School for high-school students). Talk title: 'Massive stars: their evolution and afterlife'.
- 2024 Guide assistant in *Astronomins dag - Sweden's Day and Night of Astronomy* (OSO).
- 2024 Interview to talk about working as an astronomer and moving to Sweden in Capital24.
- 2023 Comments on our press release of a Nature article for the SEE NEWS #191 (international press release here).
- 2020 Interview for the VocAr program to promote scientific careers in Argentina, available in <https://www.conicet.gov.ar/programas/vocar/recursos-publicaciones-vocar/>, *Miradas de Ciencia - Astrónom@s*.
- 2020 Short communications with local media (radio and television) to talk about the Varsavsky Prize awarded for my research.
- 2019 Divulcation talk in a guided visit for students to the IAR observatory. Talk title: 'El rompecabezas astronómico'.
- 2018 Divulcation talk in a guided visit for students to the IAR observatory. Talk title: 'La astronomía más allá de las estrellas'.
- 2018 Participation as speaker in the Series of Talks 'Viernes en el Observatorio', FCAGLP. Talk title: 'La astronomía más allá de las estrellas'.
- 2016 Interview for the program 'Dos entre comillas', FM Meliquina 99.9
- 2015 Interview for the program 'Radiofotos', FM Universidad 107.5
- 2015 Text for astronomy outreach: 'Regiones de colisión de vientos estelares', published in Bol. 50 of the Argentinian Institute of Radioastronomy.

Languages

Spanish **Mothertongue**

English **Fluent**

First Certificate Exam Grade A, 2007

Swedish **Upper-intermediate**

Completed SFI D and SVA Grundläggande delkurs 4

Publications

Full list of publications: <https://ui.adsabs.harvard.edu/public-libraries/8vGw9SXzT7u1T1IWolEw6A>

Refereed papers in international journals

1. **Constraining the intrinsic X-ray emission in Obscured AGN using ALMA millimeter-wave observations.** del Palacio, S., Behar, E., Boorman, P., König, S. & Aalto, S.
Astronomy & Astrophysics Letters, submitted, 2026.
2. **Resolving the heart of the Wolf-Rayet binary Apep with ALMA.** del Palacio, S., Vlemmings, W., Fernández-López, M., De Becker, M., et al.
Astronomy & Astrophysics, submitted, 2026.

3. **Phase-resolved X-ray spectroscopy of the Vela pulsar with *NICER* observations.** Zubieta, E, García, F., [del Palacio, S.](#), Combi, J. A., et al.
Astronomy & Astrophysics, submitted, 2026.
4. **A parsec-scale cm-to-sub-mm SED study of the compact radio jet components in NGC 1068.** Mutie, I., [del Palacio, S.](#), Beswick, R. J., Williams-Baldwin, D., et al.
Monthly Notices of the Royal Astronomical Society, accepted, 2026.
5. **The PARADIGM Project II: Characterising Nuclear and Diffuse Radio Components in Local U/LIRGs.** Lucatelli, G., Beswick, R., Moldón, J., Alberdi, A., [et al.](#)
Monthly Notices of the Royal Astronomical Society, 549, 929, 2026.
6. **Search for radio polarization in the particle-accelerating colliding-wind binaries WR 147 and HD 167971.** Blanco, A. B., De Becker, M., Benaglia, P., [del Palacio, S.](#)
Astronomy & Astrophysics, 710, 157, 2026.
7. **Upstream neutrino production and delayed jet emission in the blazar GB6 J1542+6129.** Kun, E., Hadi, B., Göblyös, A., Becker Tjus, J., [et al.](#)
Astronomy & Astrophysics, accepted, 2026.
8. **Correlation Between X-Ray and Cosmic Neutrino Sources: From Obscured AGN to Blazars.** Kun, E., Bartos, I., Ricci, C., [del Palacio, S.](#), et al.
Publications of the Astronomical Society of the Pacific, 138, 4102, 2026.
9. **Narrowband searches for continuous gravitational waves from known pulsars in the first two parts of the fourth LIGO–Virgo–KAGRA observing run.** The LIGO Scientific Collaboration, the Virgo Collaboration, the KAGRA Collaboration, [et al.](#)
The Astrophysical Journal, 1005, 221A, 2026.
10. **Bayesian parameter study of the Seyfert-starburst composite galaxies NGC 1068 and NGC 7469.** Eichmann, B., Salvatore, S., [del Palacio, S.](#), Sommani, G., et al.
Astronomy & Astrophysics, accepted, 2026.
11. **The Inner and Outer Shock Layers of Bow Shocks in Cataclysmic Variables.** Iłkiewicz, K., Knigge, C., Scaringi, S., Castro Segura, N., [del Palacio, S.](#), Veresvarska, M.
The Astrophysical Journal Letters, 1000, 22, 2026.
12. **The Millimeter/X-ray Relation in Rapidly Accreting Supermassive Black Holes at $z < 0.16$.** Venselaar, S. M., Ricci, C., [del Palacio, S.](#), Gupta, K. K., et al.
The Astrophysical Journal, 1005, 12, 2026.
13. **A Seyfert galaxy as a hidden counterpart to a neutrino-associated blazar.** Kun, E., [del Palacio, S.](#), Bartos, I., Becker Tjus, J., et al.
Nature Communications, submitted, 2026.
14. **Constraints on gravitational waves from the 2024 Vela pulsar glitch.** The LIGO Scientific Collaboration, [et al.](#)
Astronomy & Astrophysics, 1005, 12, 2026.
15. **Giant radio pulses in the magnetar XTE J1810–197 detected with the IAR’s telescopes.** Araujo Furlan, S. B., Romero, G. E., Zubieta, E., Gancio, G., [et al.](#)
The Astrophysical Journal, 710, 13, 2025.
16. **Investigating particle acceleration in the Wolf-Rayet bubble NGC 2359.** Anindya, S., Anandmayee, T., [del Palacio, S.](#), De Becker, M., et al.
Astronomy & Astrophysics, 708, 310, 2026.
17. **PSR J0537–6910: Exponential recoveries detected for 12 glitches.** Zubieta, E, Espinoza, C. M., Antonopoulou, D., Ho, W. C. G., [et al.](#)
Astronomy & Astrophysics, 706, 61, 2026.

18. **Detection of disk-jet co-precession in a tidal disruption event.** Wang, Y., Lin, Z., Wu, L., Lei, W., Wei, S., [et al.](#)
Science Advances, 11, 9068, 2025.
19. **The detection of a large-scale ionised outflow in the local CON galaxy Zw049.057.** Wethers, C. F., Aalto, S., [del Palacio, S.](#), [et al.](#)
Astronomy & Astrophysics, 704, 195, 2025.
20. **Millimeter VLBI constraints on the central magnetic field and symmetric jet production in the twin-jet galaxy NGC 1052.** Ricci, L., Baczko, A-K., Kadler, M., [et al.](#)
Astronomy & Astrophysics, 701, 214, 2025.
21. **Foreground and internal free-free absorption in particle-accelerating colliding-wind binaries.** Tasseroul, M., De Becker, Blanco, A., M., Benaglia, P., [del Palacio, S.](#)
Astronomy & Astrophysics, 701, 243, 2025.
22. **Probing the detectability of electromagnetic signatures from Galactic isolated black holes.** Martinez, J. R., Bosch-Ramon, V., Vieyro, F. L., [del Palacio, S.](#)
Astronomy & Astrophysics, 700, 49, 2025.
23. **Millimeter emission from supermassive black hole coronae.** [del Palacio, S.](#), Yang, C., Aalto, S., Ricci, C., [et al.](#)
Astronomy & Astrophysics, 701, 41, 2025.
24. **Radio study of the colliding wind binary HD 93129A near periastron and its surroundings.** Benaglia, P., [del Palacio, S.](#), Saponara, J., Blanco, A., [et al.](#)
Astronomy & Astrophysics, 698, 72, 2025.
25. **Study of the 2024 major Vela glitch at the Argentine Institute of Radioastronomy.** Zubieta, E., Missel, R., Araujo-Furlan, S., [et al.](#)
Astronomy & Astrophysics, 698, 23, 2025.
26. **A consistent radio to sub-mm pc-scale study of the nucleus of NGC 1068.** Mutie, I., [del Palacio, S.](#), Beswick, R. J., Williams-Baldwin, D., [et al.](#)
Monthly Notices of the Royal Astronomical Society, 539, 808, 2025.
27. **Search for continuous gravitational waves from known pulsars in the first part of the fourth LIGO-Virgo-KAGRA observing run.** The LIGO Scientific Collaboration, [et al.](#)
The Astrophysical Journal, 983, 99, 2025.
28. **Glitch-induced pulse profile change of PSR J0742–2822 observed from the IAR.** Zubieta, E., García, F., [del Palacio, S.](#), Espinoza, C. M., [et al.](#)
Astronomy & Astrophysics, 694 124, 2025.
29. **Unraveling the Hybrid Origins of the X-Ray Nonthermal Emission from IGR J17091–3624.** Lin, Z., Wang, Y., [del Palacio, S.](#), [et al.](#)
The Astrophysical Journal, 974, 79, 2024.
30. **Possible correlation between unabsorbed hard X-rays and neutrinos in radio-loud and radio-quiet active galactic nuclei.** Kun, E., Bartos, I., Becker Tjus, J., [et al.](#)
Physical Review D, 110, 3014, 2024.
31. **Timing irregularities and glitches from the pulsar monitoring campaign at IAR.** Zubieta, E., García, F., [del Palacio, S.](#), Araujo Furlan, S. B., [et al.](#)
Astronomy & Astrophysics, 689, 191, 2024.
32. **Joint ALMA/X-ray monitoring of the radio-quiet type 1 AGN IC 4329A.** Shablovinskaya, E., Ricci, C., Chang, C-S., Tortosa, A., [del Palacio, S.](#), [et al.](#)
Astronomy & Astrophysics, 690, 232, 2024.

33. **LOFAR Deep Fields: Probing the sub-mJy regime of polarized extragalactic sources in ELAIS-N1. I. The catalog.** Piras, S., Horellou, C., Conway, J. E., Thomasson, M., del Palacio, S., et al.
Astronomy & Astrophysics, 687, 267, 2024.
34. **Rapid dimming followed by a state transition: a study of the highly variable nuclear transient AT 2019avd over 1000+ days.** Yanan, W., Dheeraj R., P., Altamirano, D., Gurpide, A., et al.
The Astrophysical Journal, 962, 78, 2024.
35. **Probing the non-thermal physics of stellar bow shocks using radio observations.** Martínez, J. R., del Palacio, S., Bosch-Ramon, V., & Romero, G. E.
Astronomy & Astrophysics, 680, 99, 2023.
36. **The innermost jet in the hidden ultra-luminous X-ray source Cygnus X-3.** Yang, J., García, F., del Palacio, S., Spencer, R., et al.
Monthly Notices of the Royal Astronomical Society Letters, 526, 1, 2023.
37. **Search for particle acceleration in two massive Wolf-Rayet stars using uGMRT observations.** Anindya, S., Anandmayee, T., del Palacio, S., De Becker, M., et al.
Monthly Notices of the Royal Astronomical Society, 526, 750, 2023.
38. **Evidence for non-thermal X-ray emission from the double WR colliding-wind binary Apep.** del Palacio, S., García, F., De Becker, M., Altamirano, D., Bosch-Ramon, V., et al.
Astronomy & Astrophysics, 672, 109, 2023.
39. **First results of the glitching pulsars monitoring program at the Argentine Institute of Radioastronomy.** Zubietta, E., Missel, R., Sosa Fiscella, V., Lousto, C. O., del Palacio, S., et al.
Monthly Notices of the Royal Astronomical Society, 521, 4504, 2023.
40. **The radio detection and accretion properties of the peculiar nuclear transient AT 2019avd.** Yanan, W., Ranieri, D. B., del Palacio, S., Muryel, G., et al.
Monthly Notices of the Royal Astronomical Society, 520, 2417, 2023.
41. **A shared accretion instability for black holes and neutron stars.** Vincentelli, F. M., Neilsen, J., Tetarenko, A. J., Cavecchi, Y., Castro Segura, N., et al.,
Nature, 615, 45, 2022.
42. **Non-thermal emission in hyper-velocity and semi-relativistic stars.** Martínez, J. R., del Palacio, S., Bosch-Ramon, V., & Romero, G. E.
Astronomy & Astrophysics, 661, 102, 2022.
43. **The non-thermal emission from the colliding-wind binary Apep.** del Palacio, S., Benaglia, P., De Becker, M., Bosch-Ramon, V. & Romero, G. E.
Publications of the Astronomical Society of Australia, 39, 4, 2022.
44. **A persistent ultraviolet outflow from the accretion disc in a transient neutron star binary.** Castro Segura, N., Knigge, C., Long, K. S., Altamirano, D., et al.
Nature, 603, 52, 2022.
45. **Vela Pulsar: Single Pulses Analysis with Machine Learning Techniques.** Lousto, C. O., Miseel, R., Prajapati, H., Sosa Fiscella, V., López Armengol, F. G., et al.
Monthly Notices of the Royal Astronomical Society, 509, 5790, 2022.
46. **Exploring the physics behind the non-thermal emission from star-forming galaxies detected in gamma rays.** Kornecki, P., Peretti, E., del Palacio, S., Benaglia, P. & Pellizza, L. J.
Astronomy & Astrophysics, 657, 49, 2022.
47. **High-sensitivity polarimetric study of the stellar bow shock EB27.** Benaglia, P., del Palacio, S., Hales, C., & Colazo, M. E.

- Monthly Notices of the Royal Astronomical Society, 503, 251, 2021.
48. **PSR J0437–4715: The Argentine Institute of Radioastronomy 2019–2020 Observational Campaign.** Sosa Fiscella, V., del Palacio, S., Combi, L.; Lousto, C. O., et al.
The Astrophysical Journal, 908, 158, 2021.
 49. **Bow-shocks, nova shells, disc winds and tilted discs: The Nova-Like V341 Ara has it all.** Castro Segura, N., Knigge, C., Acosta-Pulido, J. A.; Altamirano, D., del Palacio, S., et al.
Monthly Notices of the Royal Astronomical Society, 501, 1951, 2021.
 50. **New insights in the HII region G18.88–0.49: hub-filament system and accreting filaments.** Dewangan, L. K., Ojha, D. K., Sharma, Saurabh, del Palacio, S., Bhadari, N. K., & Das, A.
The Astrophysical Journal, 903, 13, 2020.
 51. **The gamma-ray/infrared luminosity correlation of star-forming galaxies.** Kornecki, P., Pellizza, L. J., del Palacio, S., Müller, A. L., et al.,
Astronomy & Astrophysics, 641, 147, 2020.
 52. **Probing the physical conditions and star formation processes in the Galactic HII region S305.** Dewangan, L.K., Sharma, S., Pandey, R., del Palacio, S., Ojha, D.K., Benaglia, P., et al.
The Astrophysical Journal, 898, 172, 2020.
 53. **The high-energy emission from HD 93129A near periastron.** del Palacio, S., García, F., Altamirano, D., Barbá, R. H., Bosch-Ramon, V., Corcoran, M., et al.
Monthly Notices of the Royal Astronomical Society, 494, 6043, 2020.
 54. **Upgraded antennas for pulsar observations in the Argentine Institute of Radio astronomy.** Gancio, G., Lousto, C. O., Combi, L., del Palacio, S., Lopez Armengol, F. G., et al.
Astronomy & Astrophysics, 633, 84, 2019.
 55. **Investigating Particle Acceleration in the Wolf-Rayet Bubble G2.4+1.4.** Prajapati, P., Tej, A., del Palacio, S., Benaglia, P., et al.
The Astrophysical Journal Letters, 884, 49, 2019.
 56. **A model for high-mass microquasar jets under the influence of a strong stellar wind.** Molina, E., del Palacio, S., & Bosch-Ramon, V.
Astronomy & Astrophysics, 629, 129, 2019.
 57. **Radio observations of evaporating objects in the Cygnus OB2 region.** Isequilla, N. L., Fernández-López, M., Benaglia, P., Ishwara-Chandra, C. H., & del Palacio, S.
Astronomy & Astrophysics, 627, 58, 2019.
 58. **Investigation of the WR 11 field at decimeter wavelengths.** Benaglia, P., del Palacio, S., Ishwara-Chandra, C. H., De Becker, M., et al.
Astronomy & Astrophysics, 625, 99, 2019.
 59. **Gamma rays from jets interacting with BLR clouds in blazars.** del Palacio, S., Bosch-Ramon, & Romero, G. E.
Astronomy & Astrophysics, 623, 101, 2019.
 60. **Multi-zone non-thermal radiative model for stellar bowshocks.** del Palacio, S., Bosch-Ramon, V., Müller, A. L., & Romero, G. E.
Astronomy & Astrophysics, 617, 13, 2018.
 61. **Gamma rays from clumpy wind-jet interactions in high-mass microquasars.** de la Cita, V. M., del Palacio, S., Bosch-Ramon, V., Paredes-Fortuny, X., et al.
Astronomy & Astrophysics, 604, 39, 2017.
 62. **Synchrotron radiation and absence of linear polarization in the colliding wind binary WR 146.** Hales, C. A., Benaglia, P., del Palacio, S., Romero, G. E., & Koribalski, B. S.
Astronomy & Astrophysics, 598, 42, 2017.

63. **A model for the non-thermal emission of the very massive colliding-wind binary HD 93129A.** del Palacio, S., Bosch-Ramon, V., Romero, G. E., & Benaglia, P. *Astronomy & Astrophysics*, 591,139, 2016.
64. **Gamma-ray binaries beyond one-zone models: an application to LS 5039.** del Palacio, S., Bosch-Ramon, V., & Romero, G. E. *Astronomy & Astrophysics*, 575,112, 2015.

[Refereed papers in international journals \(non-astronomical\)](#)

1. **Comparative Study of 10 X-STR Markers in Populations of Northeast Argentina.** Di Santo Meztler, G. P., Glesmann, L. A., Esteban, M. E., del Palacio, S., et al. (*Human Biology*, 91:1-12, 2019).
2. **Genetic differentiation of North-east Argentina populations based on 30 binary X-chromosome markers.** Di Santo Meztler, G. P., del Palacio, S., Esteban, M. E., Armoa, I., et al. (*Frontiers in Genetics*, 9, 208, 2018).

[Complete papers in international proceedings](#)

1. **Linking hard X-ray and high-energy neutrino emission from radio-loud and radio-quiet AGN.** Kun, E., Bartos, I., Becker Tjus, J., et al. (Proceedings of ICRC, 2025).
2. **Monte Carlo parameter study for Seyfert AGN-starburst composite galaxies NGC1068 and NGC7469.** Salvatore, S., Eichmann, B., Sommani, G., del Palacio, S., et al. (Proceedings of ICRC, 2025).
3. **A multiscale and multi-frequency radio study of local U/LIRGs.** Lucatelli, G., Beswick, R. J., Moldón, J., et al. (Proceedings of the 16th EVN Symposium, 2024).
4. **Investigating the Role of Pre-supernova Massive Stars in the Acceleration of Galactic Cosmic Rays.** De Becker, M., del Palacio, S., Benaglia, P., Tej, A., et al. (Proceedings of the 3rd BINA Workshop on the Scientific Potential of the Indo-Belgian Cooperation, 93, 536, 2024).
5. **Study of Wolf-Rayet Stars Using uGMRT.** Saha, A., Tej, A., del Palacio, S., De Becker, M., et al. (Proceedings of the 3rd BINA Workshop on the Scientific Potential of the Indo-Belgian Cooperation, 93, 536, 2024).
6. **Prospects for Detecting Fast Transients with the Radio Telescopes of the Argentine Institute of Radio Astronomy.** Araujo Furlan, S. B., Zubieta, E., Gancio, G., Romero, G. E., et al. (*Rev. Mex. AA*, 56, 85, 2024).
7. **Updates on the glitching pulsar monitoring campaign performed from IAR** Zubieta, E., del Palacio, S., García, F., Araujo Furlan, S. B., et al. (*Rev. Mex. AA*, 56, 161, 2024).
8. **5-12 pc resolution ALMA imaging of gas and dust in the obscured compact nucleus of IRAS 17578–0400.** Yang, C., Aalto, S., König, S., del Palacio, S., Gorski, M., et al. (Proceedings of the International Astronomical Union, IAU symposium #378, 2023).
9. **Gamma-ray luminosity/star-formation rate correlations in the context of the Cherenkov Telescope Array.** Kornecki, P., Peretti, E., del Palacio, S., Marcowith, A., & Araudo, A. (Proceedings of Gamma2022, 2023).
10. **A multi-wavelength view of the cosmic ray confinement in star-forming galaxies.** Kornecki, P., Peretti, E., del Palacio, S., Benaglia, P., Pellizza, L. J., & Romero, G. E. (Proceedings of ICRC, 2021).
11. **Contribution due to inhomogeneous winds to the gamma-ray emission in microquasar jets.** de la Cita, V. M., del Palacio, S., Bosch-Ramon, V., Paredes-Fortuny, X., Romero, G. E., & Khangulyan, D. (Proceedings of Gamma2016, 2017).
12. **Gamma-ray predictions for the very massive colliding-wind binary HD 93129A.** del Palacio, S.,

- Bosch-Ramon, V., Romero, G. E., & Benaglia, P. (Proceedings of Gamma2016, 2017).
13. **Relativistic particle content in the most massive colliding-wind binary in the Galaxy.** del Palacio, S., Bosch-Ramon, V., Romero, G. E., & Benaglia, P. (Proceedings of HEPRO V, 2015).
 14. **Interactions of relativistic particles in stellar winds.** del Palacio, S., Bosch-Ramon, V., & Romero, G. E. (Proceedings of Second Argentinian-Brazilian Meeting on Gravitation, Astrophysics, and Cosmology, 2014).

Complete papers in national proceedings

1. **PuGli-S: Primeros glitches detectados desde el IAR.** Zubieta, E., del Palacio, S., García, F., Gancio, G., et al. (Bol. Asoc. Arg. Astron., 64, 268, 2023).
2. **Detección de un glitch en PSR J1048–5832 desde el IAR.** Zubieta, E., Sosa Fiscella, V., Combi, J. A., del Palacio, S., et al. (Bol. Asoc. Arg. Astron., 63, 262, 2022).
3. **Non-thermal processes in the bow shocks of hyper-velocity stars.** Martínez, J. R., del Palacio, S., & Romero, G. E. (Bol. Asoc. Arg. Astron., 62, 274, 2021).
4. **Estudio de alta precisión del púlsar J0437–4715 desde el Instituto Argentino de Radioastronomía.** Sosa Fiscella, V., del Palacio, S., Combi, L., Lousto, C. O., et al. (Bol. Asoc. Arg. Astron., 62, 268, 2021).
5. **Non-thermal radiation associated with massive stars.** del Palacio, S., Bosch-Ramon, V., & Romero, G. E. (Bol. Asoc. Arg. Astron., 62, 246, 2021).
6. **Towards a radio-polarimetric study of the bright colliding-wind binary WR 147.** Blanco, A., Benaglia, P., del Palacio, S., & Hales, C. A. (Bol. Asoc. Arg. Astron., 61, 66, 2020).
7. **Gamma-ray emission from interactions between jets and BLR clouds.** del Palacio, S., Bosch-Ramon, V., & Romero, G. E. (Bol. Asoc. Arg. Astron., 60, 92, 2018).
8. **A model for the non-thermal emission of the very massive colliding-wind binary HD93129A.** del Palacio, S., Bosch-Ramon, V., Romero, G. E., & Benaglia, P. (Bol. Asoc. Arg. Astron., 58, 225, 2016).
9. **High-Energy Radiative Processes in High-Mass Binary Systems.** del Palacio, S., Bosch-Ramon, V., & Romero, G. E. (Bol. Asoc. Arg. Astron. 56, 355, 2013).

Institutional Participation

- 2024–Present Organiser of the weekly astronomy seminars at Chalmers University.
- 2025 Postdoc representative for the advice panel on the merger of the Astronomy division and the Physics department at Chalmers University of Technology.
- 2015–2018 Graduate member of the Astronomy Curriculum Reform Comission of the Faculty of Astronomical and Geophysical Sciences of La Plata.
- 2017 Graduate member of the Advisory Committee for selecting a graduate teaching assistant for Calculus, Faculty of Engineering of La Plata. File 300-9203/16.
- 2011 Student member of the Advisory Committee for selecting the student teaching assistant for General Astronomy, Faculty of Astronomical and Geophysical Sciences of La Plata. File 1100-1387.

Participation in Research Groups and Scientific Associations

- 2023–Present Member of the CONquest research group.
- 2023–Present Member of the *European Astronomical Society (EAS)*.

- 2021–Present Member of the collaboration *Particle Acceleration and Non-Thermal Emission of Radiation in Astrophysics - Stars (PANTERA-Stars)*. Webpage: <http://www.astro.ulg.ac.be/~debecker/pantera/>.
- 2019–2023 Participant of the International Exchanges of the Royal Society *The first step for High-Energy Astrophysics relations between Argentina and UK*. Webpage: <http://rayosx.fcaglp.unlp.edu.ar/soton/index.html>
- 2017–Present Member of the group *Pulsar Monitoring in Argentina (PuMA)*. Webpage: <http://puma.iar.unlp.edu.ar/>.
- 2015–Present Member of the group *Grupo de Astrofísica Relativista y Radioastronomía (GARRA)*. Webpage: <https://garra.iar.unlp.edu.ar/>.
- 2015–2023 Professional partner of the *Asociación Argentina de Astronomía (AAA)*.

Participation as reviewer of scientific articles

Reviewer of 1 article in ApJL, 1 article in ApJ, 2 articles in MNRAS, 3 proceedings in the BAAA, and 1 proceeding in the Proceedings of Science (Gamma2022).

Participation as reviewer of observational proposals

Reviewer in a TAC for *XMM-Newton* AO25. Reviewer of ALMA proposals in cycles 9, 10 and 11 (2022–2024).

Contact references

- (PhD advisor) **Dr. Valentí Bosch-Ramon**. Assoc. Prof. at Barcelona University, Correspondent Researcher of CONICET. Contact: vbosch@fqa.ub.edu
- (Line manager) **Dr. Matthias Maercker**. Researcher at Chalmers University. Contact: matthias.maercker@chalmers.se
- Dr. Susanne Aalto**. Professor and Deputy President at Chalmers University. Contact: susanne.aalto@chalmers.se